

Exercises

- 1.⁻ Determine the Ramsey number $R(3)$.
- 2.⁻ Deduce the case $k = 2$ (but c arbitrary) of Theorem 9.1.3 directly from Theorem 9.1.1.
3. Prove that the graphs not containing some fixed star as an induced subgraph are χ -bounded.
(See Chapter 5.6 for the definition of χ -boundedness.)
4. Let the edges of a complete graph G on $\mathbb{N}$ be coloured with two colours. Show that G contains two monochromatic paths whose vertex sets partition $\mathbb{N}$.
5. An *arithmetic progression* is an increasing sequence of numbers of the form $a, a + d, a + 2d, a + 3d \dots$. *Van der Waerden's theorem* says that no matter how we partition the natural numbers into two classes, one of these classes will contain arbitrarily long arithmetic progressions. Must there even be an infinite arithmetic progression in one of the classes?
6. Can you improve the exponential upper bound on the Ramsey number $R(n)$ for perfect graphs?
7. Show that the n^ϵ -type bound in the Erdős-Hajnal conjecture cannot be lowered substantially, by finding a graph H such that infinitely many graphs G not containing H as an induced subgraph satisfy $\alpha(G) \leq |G|^{1/2}$ as well as $\omega(G) \leq |G|^{1/2}$.
8. Let $\mathcal{G}$ be a class of graphs that is closed under taking induced subgraphs. Assume that $\mathcal{G}$ is χ -bounded, with a polynomial χ -bounding function. Prove the Erdős-Hajnal conjecture for the graphs in $\mathcal{G}$.
- 9.⁺ Construct a graph on $\mathbb{R}$ that has neither a complete nor an edgeless induced subgraph on $|\mathbb{R}| = 2^{\aleph_0}$ vertices. (So Ramsey's theorem does not extend to uncountable sets.)
10. (i)⁻ Use Ramsey's theorem to show that for any $k, \ell \in \mathbb{N}$ there is an $n \in \mathbb{N}$ such that every sequence of n distinct integers has an increasing subsequence of length $k + 1$ or a decreasing subsequence of length $\ell + 1$.
(ii) Prove that $n = k\ell + 1$ has this property but $n = k\ell$ does not.
- 11.⁺ Find a short proof of the Erdős-Pósa theorem (2.3.2) with a recursively defined bound.
(Hint. For the recursion step find two cycles with at most one vertex in common. If Menger's theorem between them does not give you k disjoint cycles, use the size of the separator it offers for the definition of $f(k)$ from $f(k - 1)$.)
- 12.⁺ Prove the edge version of the Erdős-Pósa theorem (2.3.2): there exists a function $g: \mathbb{N} \rightarrow \mathbb{R}$ such that, given $k \in \mathbb{N}$, every graph contains either k edge-disjoint cycles or a set of at most $g(k)$ edges meeting all its cycles.
(Hint. Consider in each component a normal spanning tree.)

13. Show that for every $k \in \mathbb{N}$ there is an $n \in \mathbb{N}$ such that among any n points in the plane, no three of them collinear, there are k points spanning a convex k -gon, i.e. such that none of them lies in the convex hull of the others.
14. Show that for every $k \in \mathbb{N}$ there is an $n \in \mathbb{N}$ such that, for every partition of $\{1, \dots, n\}$ into k sets, at least one of the subsets contains numbers x, y, z such that $x + y = z$.
15. Let $(X, \leq)$ be a totally ordered set, and let $G = (V, E)$ be the graph on $V := [X]^2$ with $E := \{(x, y)(x', y') \mid x < y = x' < y'\}$.
 - (i) Show that G contains no triangle.
 - (ii) Show that $\chi(G)$ will get arbitrarily large if $|X|$ is chosen large enough.
16. A family of sets is called a Δ -system if every two of the sets have the same intersection. Show that every infinite family of sets of the same finite cardinality contains an infinite Δ -system.
17. Prove that for every $r \in \mathbb{N}$ and every tree T there exists a $k \in \mathbb{N}$ such that every graph G with $\chi(G) \geq k$ and $\omega(G) < r$ contains a subdivision of T in which no two branch vertices are adjacent in G (unless they are adjacent in T).
18. Let $m, n \in \mathbb{N}$, and assume that $m - 1$ divides $n - 1$. Show that every tree T of order m satisfies $R(T, K_{1,n}) = m + n - 1$.
19. Prove that $2^c < R(2, c, 3) \leq 3c!$ for every $c \in \mathbb{N}$.
(Hint. Induction on c .)
20. Explain why, in the proof of Theorem 9.2.2, choosing ϵ small enough can ensure that the regularity graph R contains a copy of K^ϵ , although some of the pairs (V_i, V_j) in G may not be ϵ -regular. Your explanation may use that $t_{\ell-1}(k) \approx \frac{\ell-2}{\ell-1} \binom{k}{2}$, but should contain no calculations.
- 21.⁺ Determine the Ramsey-minimal graphs for stars.
22. Prove the analogue of Theorem 9.3.1 for vertex colouring: for every graph H there exists a graph G which, given any 2-colouring of its vertices, contains a monochromatic induced copy of H .
23. Show that the Ramsey graph G for H constructed in the proof of Theorem 9.3.1 satisfies $\omega(G) = \omega(H)$.
24. In the proof of Theorem 9.3.1, is it really necessary to equip G^{k+1} for $i \notin \{i_1, i_2\}$ with separate disjoint copies of V_k^i , one for every p , or could we define G^{k+1} from G^k by just replacing P with P' and joining it to the other V_i^k in the right way?
- 25.⁻ Show that any Kuratowski set for a non-trivial graph property is unique up to equivalence.

26. Deduce Theorem 9.4.5 (iii) from Theorem 9.4.3, and vice versa.

Notes

Due to increased interaction with research on random and pseudo-random⁵ structures (the latter being provided, for example, by the regularity lemma), the Ramsey theory of graphs has recently seen a period of major activity and advance. Theorem 9.2.2 is an early example of this development.

For the more classical approach, the introductory text by R.L. Graham, B.L. Rothschild & J.H. Spencer, *Ramsey Theory* (2nd edn.), Wiley 1990, makes stimulating reading. This book includes a chapter on graph Ramsey theory, but is not confined to it. Surveys of finite and infinite Ramsey theory are given by J. Nešetřil and A. Hajnal in their chapters in the *Handbook of Combinatorics* (R.L. Graham, M. Grötschel & L. Lovász, eds.), North-Holland 1995. The Ramsey theory of infinite sets forms a substantial part of combinatorial set theory, and is treated in depth in P. Erdős, A. Hajnal, A. Máté & R. Rado, *Combinatorial Set Theory*, North-Holland 1984. An attractive collection of highlights from various branches of Ramsey theory, including applications in algebra, geometry and point-set topology, is offered in B. Bollobás, *Graph Theory*, Springer GTM 63, 1979.

Ramsey's original theorem, Theorem 9.1.1, is from F.P. Ramsey, On a problem of formal logic, *Proc. Lond. Math. Soc.* **2** (1930), 264–286. Our proof is due to P. Erdős and G. Szekeres, A combinatorial problem in geometry, *Compos. Math.* **2** (1935), 463–470. Its bound of $R(r) \leq 2^{2^{r-3}} = 4^r/8$ was improved by an exponential factor, to $(4 - \epsilon)^r$ for some $\epsilon > 0$, only as recently as in 2023, by M. Campos, S. Griffiths, R. Morris and J. Sahasrabudhe, An exponential improvement for diagonal Ramsey, arXiv:2303.09521.

The Erdős-Hajnal conjecture is taken from P. Erdős & A. Hajnal, Ramsey-type theorems, *Discrete Appl. Math.* **25** (1989), 37–52. A survey on the state of the art a couple of years ago was given by M. Chudnovsky, The Erdős-Hajnal conjecture – a survey, *J. Graph Theory* **75** (2014), 178–190, arXiv:1606.08827.

Theorem 9.2.2 is due to V. Chvátal, V. Rödl, E. Szemerédi & W.T. Trotter, The Ramsey number of a graph with bounded maximum degree, *J. Comb. Theory, Ser. B* **34** (1983), 239–243. The theorem marked a breakthrough towards a conjecture of Burr and Erdős (1975), which asserts that the Ramsey numbers of graphs with bounded average degree in every subgraph are linear: for every $d \in \mathbb{N}$, the conjecture says, there exists a constant c such that $R(H) \leq c|H|$ for all graphs H with $d(H') \leq d$ for all $H' \subseteq H$. This conjecture was eventually proved by C. Lee, Ramsey numbers of degenerate graphs, *Ann. Math.* **185** (2017), 791–829, arXiv:1505.04773.

Our proof of Theorem 9.3.1 is from J. Nešetřil & V. Rödl, Simple proof of the existence of restricted Ramsey graphs by means of a partite construction, *Combinatorica* **1** (1981), 199–202. This construction technique was refined in an iterative and highly complex way for the proof of Theorem 9.3.4 in

⁵ Concrete graphs whose structure resembles the structure expected of a random graph are called *pseudo-random*. For example, the bipartite graphs spanned by an ϵ -regular pair of vertex sets in a graph are pseudo-random.